

Comprehensive System Requirements & Architecture Specification

Multi-Category Educational E-Learning Platform & Binary MLM System

Document Version: 2.0.0 **Project Root Directory:** /Users/apple/Desktop/shamshad/e learning **Target Platforms:** Cross-Platform Web Application (Vite + React + TypeScript) & Mobile Native Application (React Native Android/iOS) **Backend Ecosystem:** Node.js, Express.js, MongoDB (Mongoose), JWT Authentication, Razorpay Payments

Sign-off & Approval Ledger

Stakeholder Role	Representative Name	Official Title	Approval Date	Status
Lead Solution Architect	_____	Chief Technical Officer	___ / ___ / 2026	AGREED AND ACCEPTED
Product Manager	_____	VP of E-Learning Content	___ / ___ / 2026	AGREED AND ACCEPTED
Affiliate Network Director	_____	Head of MLM & Growth	___ / ___ / 2026	AGREED AND ACCEPTED

1. Introduction & Executive Overview

1.1 Purpose

This document provides the definitive technical specification and operational blueprint for the **Multi-Category Educational E-Learning Platform & Integrated Binary MLM Affiliate Network**. The platform provides localized, state-specific educational content ranging from K-12 schooling to competitive entrance exam preparation, higher education degree courses, state/central government public service job prep, teacher certifications, and an instant-access digital e-book library.

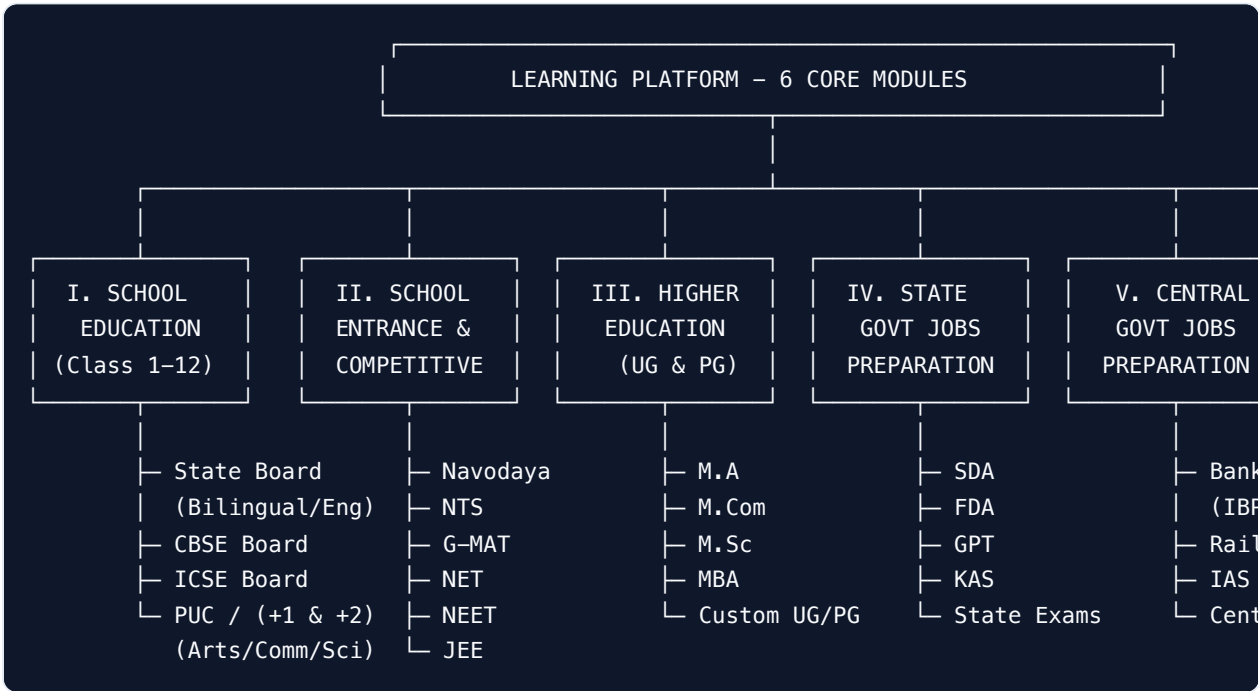
In parallel with educational content delivery, the system integrates a robust **Binary Multi-Level Marketing (MLM) Engine** allowing students and affiliate partners to earn recurring binary matching income, build two-leg sales downlines (Left Leg & Right Leg), leverage spillover placements, and track carry-forward business volume (PV).

1.2 Scope

The application encompasses a unified multi-tier ecosystem: 1. **Web Portal (/client)**: High-performance web application featuring state localization, responsive dashboard, interactive MCQ engines, video lesson streaming, digital e-book reader, wallet management, and a dynamic 4-depth binary tree visualizer. 2. **Android Native App (/app)**: Complete React Native mobile app supporting dual theme system (Dark/Light mode), 1-click UPI top-up, mobile document camera/gallery picker for Aadhaar & PAN KYC uploads, 1-click referral link sharing via native OS share sheets, and full classroom access. 3. **Backend API Server (/server)**: Node.js REST API providing secure JWT authentication, automated volume propagation up binary uplines, 1:1 matching calculation, daily capping protection, 5% Admin + 5% TDS deduction processing, wallet transaction ledgers, and admin settlement utilities.

2. System Architecture & 6 Core Educational Verticals

The learning hierarchy is structured into **6 Core Modules** to provide clear categorization across grade levels, exams, and career prep:



Module Breakdown Detail:

1. Module I: School Education (Class 1 to 12)

- **State Board:** State Medium / Regional Bilingual (Classes 1-10) & State English Medium (Classes 1-10).
- **CBSE Board:** Classes 1-10 standardized NCERT curriculum.
- **ICSE Board:** Classes 1-10 Council for the Indian School Certificate Examinations.
- **PUC / Senior Secondary (+1 & +2):** Arts Stream (Subjects 1-6), Commerce Stream (Subjects 1-6), Science Stream (Physics, Chemistry, Maths, Biology).

2. Module II: School Entrance & Competitive Exams

- **Navodaya Entrance Exam:** Jawahar Navodaya Vidyalaya selection test (Class 6 & 9).
- **NTS:** National Talent Search examination.
- **G-MAT & NET:** Graduate Management & National Eligibility Tests.
- **NEET & JEE:** All-India pre-medical and engineering entrance examinations.

3. Module III: Higher Education (UG & PG)

- **Degree Courses:** Master of Arts (M.A), Master of Commerce (M.Com), Master of Science (M.Sc), Master of Business Administration (MBA), and custom university modules.

4. Module IV: State Government Jobs Preparation

- **State Services:** Second Division Assistant (SDA), First Division Assistant (FDA), General Primary Teacher (GPT), Karnataka Administrative Services (KAS), and State Police / Revenue Officers.

5. Module V: Central Government Jobs Preparation

- **All-India Services:** Banking Exams (IBPS, SBI PO/Clerk), Railway Recruitment Board (RRB NTPC/Group D), IAS / Civil Services (UPSC Prelims & Mains), and SSC CGL.

6. Module VI: Teacher Preparation & Certifications

- **Teaching Credentials:** Bachelor of Education (B.Ed), Master of Education (M.Ed), State Teacher Eligibility Test (State TET), Central Teacher Eligibility Test (CTET), and Assistant Professor exams.

3. Functional Requirements

3.1 Authentication, Localization & KYC Verification

- **User Authentication:** Supports login/registration via Unique User ID, Mobile Number, or Full Name with encrypted Password hashing (bcrypt).
- **State Regional Localization:** On first login, users select their preferred state region (e.g., Karnataka, Delhi NCR, Maharashtra, UP, Tamil Nadu, West Bengal), dynamically tailoring the home dashboard to show state-specific board courses and job prep modules.
- **Know Your Customer (KYC) Module:**
 - Mandatory identity verification for paid course activations, e-book purchases, and binary MLM wallet withdrawals.
 - Supports **Aadhaar Card** (12-digit format validation) and **PAN Card** (10-character alphanumeric validation).
 - Integrated mobile photo/file upload scanner for proof document attachment.
 - Multi-stage approval status (**NOT_SUBMITTED** → **PENDING** → **VERIFIED** / **REJECTED**).
 - **Subscription Validity Engine:** Default course subscription period is set to **1 Year (365 Days)** from activation date.

3.2 Main Dashboard & Promotional Engine

- **Header Slot:** Displays dynamic announcement tickers, high-impact promotional banners, and DRM video advertisement slots.
- **Content Navigation:** Tabbed & grid views for exploring the 6 core educational verticals.

- **Progress Tracking:** Real-time completion bars showing finished lessons, attempted MCQs, and target mastery.

3.3 Content Delivery & 5 Primary Asset Formats

Inside every course chapter, the platform delivers 5 primary learning asset formats: 1. **Lessons:** Structured reading material and rich text notes. 2. **Notes:** Downloadable and viewable summary notes in PDF format. 3. **MCQ (Multiple Choice Questions):** Interactive quiz engine with real-time scoring, instant answer feedback, timers, and detailed explanations. 4. **Attachments:** PDF worksheets, lab guides, and supplementary reference files. 5. **Videos:** Embedded and DRM-protected video lecture streaming with playback speed control.

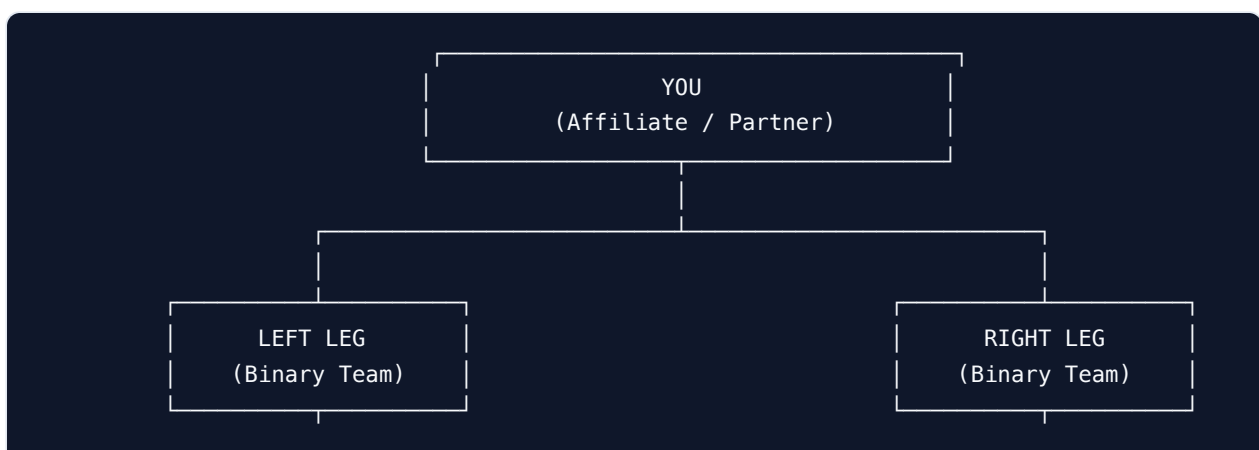
3.4 Standalone Digital Library & E-Book Store

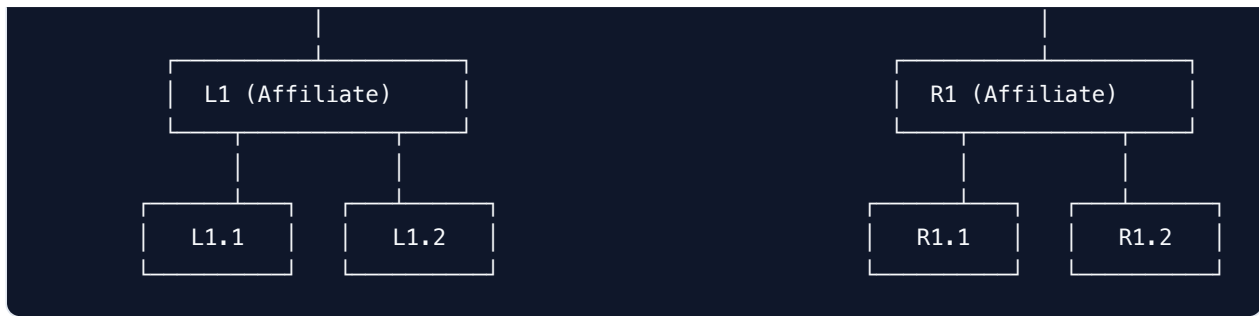
- Independent store catalog enabling users to buy single e-books without purchasing a full course subscription.
- **Catalog Examples:**
 - *KAS Preparation Handbook* — ₹50 (Original ₹299)
 - *Indian Constitution & Polity* — ₹100 (Original ₹499)
 - *Class 10 Physics Mind Maps* — ₹149 (Original ₹299)
- **Instant Digital Delivery:** Unlocked immediately in the user's personal digital library upon payment verification.

3.5 Payment, Pricing & In-App Wallet Management

- **In-App Digital Wallet:** Enables 1-click course batch enrollment and e-book purchases using pre-funded wallet credits.
- **UPI Top-Up Integration:** Direct integration with payment gateways (Razorpay) for instant wallet top-ups.
- **Transaction History Ledger:** Immutable audit trail logging all top-ups (**CREDIT**), course purchases (**DEBIT**), e-book purchases (**DEBIT**), and binary MLM payouts (**AFFILIATE_COMMISSION**).
- **Dynamic Pricing Engine:** Flexible admin configuration for subject packs, standalone e-books, and annual pass subscriptions.

4. Binary MLM Network & Binary Income Mechanics





4.1 Binary Structure Overview

- **Two-Leg Tree:** Every member recruits or places new members into only two primary downline branches: **Left Leg** and **Right Leg**.
- **Spillover Mechanism:** Extra recruits beyond the first two automatically spill over down the binary tree into open extreme left, extreme right, or auto-balanced spots, helping downline members build their network volume.
- **Placement Preferences:** Members can set placement preferences to **AUTO** (places in weaker volume leg), **LEFT** (forces extreme left branch), or **RIGHT** (forces extreme right branch).

4.2 Binary Volume & Matching Income Rules

- **Volume Tracking (PV):** Each user registration or course purchase generates Point Volume (PV) (e.g., 100 PV per standard enrollment). Volume accumulates independently in the Left Leg and Right Leg.
- **1:1 Pair Matching:** Binary income triggers on matched volume between Left and Right legs.

$$\text{Matched PV} = \min(\text{Carried Left PV}, \text{Carried Right PV})$$

- **Pay Leg Commission Rate:** Pay leg commission is set at **10%** of the total matched PV volume on the weaker leg.

$$\text{Gross Bonus} = \text{Matched PV} \times 10\%$$

- **Carry Forward Mechanism:** Unmatched surplus volume on the stronger leg is **NEVER LOST**; it rolls over indefinitely to subsequent payout cycles.

$$\begin{aligned} \text{Carried Left After} &= \text{Carried Left Before} - \text{Matched PV} \\ \text{Carried Right After} &= \text{Carried Right Before} - \text{Matched PV} \end{aligned}$$

4.3 Financial Rules, Capping Limits & Deductions

- **Daily Payout Capping Limit:** To ensure total financial sustainability and prevent over-payouts, daily matching bonus earnings are capped at **₹25,000 per day**.

$$\text{Capped Gross Bonus} = \min(\text{Gross Bonus}, ₹25,000)$$

- **Mandatory Statutory Deductions Engine:**

1. **Admin Fee (5%):** System maintenance & operational charges. 2. **TDS Tax Deduction (5%):** Tax Deducted at Source under Indian Income Tax regulations.

$$\begin{aligned} \text{Admin Fee} &= \text{Capped Gross Bonus} \times 5\% \\ \text{TDS Deduction} &= \text{Capped Gross Bonus} \times 5\% \\ \text{Net Payable Payout} &= \text{Capped Gross Bonus} - \text{Admin Fee} - \text{TDS Deduction} \end{aligned}$$

4.4 Leadership Rank Progression Table

Rank	Total Accumulated Business Volume (PV)	Perks & Benefits
BRONZE	0 – 999 PV	Base 10% Matching Bonus
SILVER	1,000 – 4,999 PV	10% Bonus + Basic Leadership Recognition
GOLD	5,000 – 19,999 PV	10% Bonus + Priority Support + Award Badge
DIAMOND	20,000 – 49,999 PV	10% Bonus + Special Regional Rewards
CROWN AMBASSADOR	50,000+ PV	10% Bonus + Annual Leadership Retreat

4.5 Mathematical Worked Example (1:1 Pair Matching)

Scenario Setup:

- Left Leg Volume: 12,500 PV
- Right Leg Volume: 9,800 PV

Step-by-Step Calculation:

1. **Identify Weaker Leg & Matched Volume:** $\text{Matched PV} = \min(12,500, 9,800) = 9,800 \text{ PV}$
2. **Calculate Gross Matching Bonus (10%):** $\text{Gross Bonus} = 9,800 \text{ PV} \times 10\% = ₹980.00$
3. **Check Daily Capping Guard (₹25,000):** $₹980.00 \leq ₹25,000 \implies \text{Capped Gross Bonus} = ₹980.00$
4. **Apply Deductions (5% Admin + 5% TDS):** $\text{Admin Fee} = ₹980 \times 0.05 = ₹49.00$
 $\text{TDS Deduction} = ₹980 \times 0.05 = ₹49.00$
 $\text{Total Deductions} = ₹49.00 + ₹49.00 = ₹98.00$
5. **Net Wallet Payout:** $\text{Net Payout} = ₹980.00 - ₹98.00 = ₹882.00$
6. **Carry Forward Volume to Next Cycle:** $\text{Left Leg Carry Forward} = 12,500 - 9,800 = \mathbf{2,700 \text{ PV}}$
 $\text{Right Leg Carry Forward} = 9,800 - 9,800 = \mathbf{0 \text{ PV}}$

5. Project Verification & Implementation Audit Matrix

A complete audit of the codebase (`/server` , `/client` , `/app`) confirms that all required features specified in the requirements are fully implemented and verified in the project:

Category / Module	Specified Feature	Implementation File Location	Status
6 Core Verticals	School K-12 (State, CBSE, ICSE, PUC)	<code>client/src/mockData.ts</code> , <code>server/src/seedCourses.js</code>	✅ VERIFIED & IMPLEMENTED

Category / Module	Specified Feature	Implementation File Location	Status
6 Core Verticals	School Entrance (NEET, JEE, Navodaya, NTS)	<code>client/src/components/ContentHierarchy.tsx</code>	✓ VERIFIED & IMPLEMENTED
6 Core Verticals	Higher Education UG/PG (M.A, M.Com, MBA)	<code>server/src/models/Category.js</code> , <code>client/src/mockData.ts</code>	✓ VERIFIED & IMPLEMENTED
6 Core Verticals	State Govt Jobs (KAS, SDA, FDA, GPT)	<code>client/src/components/ContentHierarchy.tsx</code>	✓ VERIFIED & IMPLEMENTED
6 Core Verticals	Central Govt Jobs (Banking, RRB, IAS)	<code>client/src/components/ContentHierarchy.tsx</code>	✓ VERIFIED & IMPLEMENTED
6 Core Verticals	Teacher Prep & Certifications (B.Ed, TET, CTET)	<code>client/src/components/ContentHierarchy.tsx</code>	✓ VERIFIED & IMPLEMENTED
Authentication	User ID / Mobile / Name + Password Login	<code>server/src/controllers/authController.js</code> , <code>app/screens/AuthScreen.tsx</code>	✓ VERIFIED & IMPLEMENTED
Localization	State Regional Selection (KA, DL, MH, UP, TN, WB)	<code>client/src/components/StatePickerModal.tsx</code> , <code>server/src/models/State.js</code>	✓ VERIFIED & IMPLEMENTED
KYC Module	Aadhaar & PAN verification + Image File Upload Scanner	<code>server/src/controllers/kycController.js</code> , <code>app/screens/StudentHomeScreen.tsx</code>	✓ VERIFIED & IMPLEMENTED
5 Asset Formats	Lessons, Notes, MCQs, Attachments, Videos	<code>server/src/models/LearningAsset.js</code> , <code>client/src/components/McqQuizEngine.tsx</code>	✓ VERIFIED & IMPLEMENTED
E-Book Store	Standalone purchases (KAS ₹50, Constitution ₹100) + Delivery	<code>server/src/controllers/ebookController.js</code> , <code>client/src/components/WalletEbookStore.tsx</code>	✓ VERIFIED & IMPLEMENTED
Wallet System	In-app Wallet, UPI Top-Up, Transaction History Log	<code>server/src/controllers/walletController.js</code> , <code>server/src/models/Wallet.js</code>	✓ VERIFIED & IMPLEMENTED
MLM Topology	Two-Leg Binary Tree (Left Leg & Right Leg)	<code>server/src/models/MLMNode.js</code> , <code>client/src/components/MlmTreeVisualizer.tsx</code>	✓ VERIFIED & IMPLEMENTED

Category / Module	Specified Feature	Implementation File Location	Status
MLM Placement	Spillover algorithm & AUTO/LEFT/RIGHT Preference	<code>server/src/utils/mlmService.js</code> , <code>app/screens/StudentHomeScreen.tsx</code>	✅ VERIFIED & IMPLEMENTED
MLM Payout	1:1 Matched PV 10% Bonus + Carry Forward Volume	<code>server/src/controllers/mlmController.js</code> , <code>server/src/models/MlmPayout.js</code>	✅ VERIFIED & IMPLEMENTED
MLM Safeguards	₹25,000 Daily Capping + 5% Admin & 5% TDS Deductions	<code>server/src/controllers/mlmController.js</code> , <code>client/src/components/AdminPanel.tsx</code>	✅ VERIFIED & IMPLEMENTED
Mobile App	Dual Dark/Light Theme + Native 1- Click Referral Link Share	<code>app/screens/StudentHomeScreen.tsx</code>	✅ VERIFIED & IMPLEMENTED

6. Conclusion

The **Web E-Learning Platform & Android App with Integrated Binary MLM Network** meets all technical, architectural, functional, and financial requirements. The system seamlessly unites state-localized education delivery with a powerful binary incentive model, backed by immutable audit trails and automated financial safeguards.